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Joanna Gutierrez

Education

University of Texas Rio Grande Valley (UTRGV) Expected May 2027
B.S., Computer Science, Minor in Electrical Engineering
Overall GPA: 3.61/4.0, Major GPA: 3.92/4.0

Publications

P. Li, ***J. Gutierrez****, F. Hernandez, Q. Lu, L. Zhou.* "LLM-Foraging: Large Language Models for Decentralized Swarm Robot Foraging." To appear in Proceedings of Distributed Autonomous Robotic Systems (DARS 2026), Oct. 2026, Tokyo, Japan. [Accepted]
Preprint: `https://arxiv.org/abs/2605.01461`

J. Becerril, M. Vargas, J. Herrera, ***J. Gutierrez****, J. Rios, M. Amjadian, C. Tarawneh, J. Yang, Q. Lu.* "Non-Contact Vibration-Based Damage Detection of Civil Structures Using a Cost-Effective Autonomous UAV." In Proceedings of International Conference on Unmanned Aircraft Systems (ICUAS 2026), June 2026, Corfu, Greece. `https://doi.org/10.1109/ICUAS69441.2026.11598690`

Research Experience

DUB REU, University of Washington June. 2026 – Aug. 2026
Make4All Lab
Advisors: Kate Glazko, Jennifer Mankoff

- Designed a LangGraph-based agentic workflow that generates fabrication-ready STL & SVG files from design images, used in a BLV (blind & low-vision) hands-on workshop
- Web-scraped 2,000+ fabrication-related posts across 16 Reddit communities & Bambu Lab Community Forum (BeautifulSoup, PlayWright)
- Qualitatively analyzed 300+ posts regarding AI-use for fabrication, identifying emerging themes related to AI-assisted design & troubleshooting

Undergraduate Researcher, UTRGV Oct. 2025 – Aug. 2026
Multiple Autonomous Robotic Systems Lab
Advisor: Qi Lu

- Developed & trained a custom genetic algorithm (GA) with Python, ROS2, & Gazebo to find an optimal set of parameters for the Central Place Foraging Algorithm (CPFA)
- Created a Gazebo, ROS2/Ardupilot SITL (Software In The Loop) development environment for natural-language driven object detection using YOLOWorld & ChatGPT
- Supported development, data collection & experimentation of a UAV-based NDT (non-destructive testing) method for structural vibration anomalies

Robots In The Real World REU, Oregon State University June. 2025 – Aug. 2025
Social Haptics & Embodiment Lab
Advisors: Antara Shah, Naomi Fitter

- Extended & prototyped a multi-modal (touch & speech) trivia interface for older adults with Python/QThread/PyQt; designed robot behaviors & audio cues (JSON, AWS Polly)
- Co-led a pilot study with 8 user-studies for a robot-hosted trivia activity with varying levels of difficult & reward
- Analyzed accuracy & response-time data to investigate the effects of reward & difficulty level on older-adult engagement & self-efficacy

Teaching Experience

Teaching Assistant

Data Structures & Algorithms – Qi Lu (3333-01)	Fall 2026
Data Mining – Li Zhang (6366-01)	Spring 2026
Discrete Mathematics – Li Zhang (3310-01, 3310-04, 3310-05)	Fall 2025 – Spring 2026

Peer Led Team Leader

Calculus 1 – Joseph O. Alanis (2413-07P)	Fall 2024
Calculus 1 – Rodrigo H. Pena Jr. (2413-09P)	Spring 2024

Fellowships & Awards

Break Through Tech (BTT) AI/ML Fellow May. 2026 – Present

- Selected as 1/1k fellows nationwide out of 4k+ applicants for a highly competitive year long AI/ML program
- Awarded fully-funded enrollment in eCornell's Machine Learning Foundations certificate program by Cornell University
- Constructing a CRDC (Civil Rights Data Collection) 2017-2024 data-model for a ML-driven dashboard that provides seclusion & restriction risk assessments to school-administrators (*Advised by Caroline Bell, Capital One*)

Keppel AmFels Scholarship Recipient Fall 2026

Robert L. Townsend Scholarship Recipient Fall 2026

Stars Scholarship Recipient Fall 2025

Last Mile REU Fund Recipient Summer 2025

Technical Skills

Programming Python, C++, C, Dart, R, SQL

Data Analysis NumPy, Pandas, Matplotlib, SciPy, Seaborn,

Robotics & AI Frameworks ROS2, Scikit-Learn, Keras, LangChain, LangGraph

Simulation & Tools Gazebo, RViz, Git\Github, Docker, Linux, Jamovi, Jupyter Notebook, Google Colab